



# DAYANANDA SAGAR UNIVERSITY

## SCHOOL OF ENGINEERING



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### DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING (ARTIFICIAL INTELLIGENCE & MACHINE LEARNING)

### AI-BASED PLAGIARISM DETECTION AND AI CONTENT ANALYSIS SYSTEM

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## INTRODUCTION

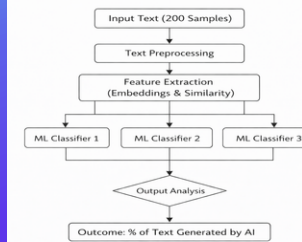
- Plagiarism is a major issue in academics and digital content creation.
- Traditional detection methods are often limited to exact text matching.
- An AI-Based Plagiarism Detection System uses Artificial Intelligence and Natural Language Processing.
- It detects both direct copying and paraphrased content.
- The system ensures accuracy, efficiency, and originality.

## OBJECTIVES

- To develop an intelligent system that can detect plagiarism by identifying both exact copying and paraphrased content using advanced NLP techniques.
- To incorporate machine learning models that analyze text patterns, semantic similarity, and embeddings to accurately classify human-written, AI-generated, and modified content.
- To design a scalable and user-friendly system that supports continuous updates, enabling improved detection performance with evolving AI-generated content.

## METHODOLOGY

- Collect and preprocess text data from various sources including human-written, AI-generated, and paraphrased content. Perform cleaning, tokenization, normalization, and removal of stop words to ensure high-quality input for analysis.
- Extract important textual features such as word embeddings, semantic similarity, and syntactic patterns to understand relationships between different texts and prepare data for model training.
- Apply machine learning models to classify the text into categories like human-written, AI-generated, or paraphrased by analyzing patterns and similarities in the extracted features.
- Compare the processed text with existing datasets using similarity measures and evaluate model performance using metrics like accuracy, precision, and recall to ensure reliable detection results.
- Integrate all modules into a unified system that provides users with an easy-to-use interface and generates clear, percentage-based reports for plagiarism detection and content authenticity.



## ARCHITECTURE

- Prepare and preprocess input text data by performing cleaning, tokenization, normalization, and noise removal, followed by feature extraction such as word embeddings, semantic similarity, and syntactic patterns for analysis.
- Apply machine learning models to classify text (human, AI, paraphrased) and compare it with existing datasets using similarity measures, then present results through a user-friendly interface with percentage-based outputs.
- Output result showing text category and percentage.

## RESULT AND CONCLUSION

- No single feature can fully detect plagiarism.
- Data shows distinct patterns between human, AI, and paraphrased text.
- Model identifies three outputs: human (0), AI (1), paraphrased (2).
- System improves plagiarism detection using AI techniques.
- Accuracy depends on proper feature selection and training data.
- Requires continuous updates and ensures data privacy.

## REFERENCES

- [1] Baly, F., Mishra, A., & Ekstrand, M. D. (2018). Detecting neural text generation: A benchmark for open domain dialogue systems. In Proceedings of the 27th ACM International Conference on Information and Knowledge Management (pp. 287-296). Link][<https://dl.acm.org/doi/10.1145/3269206.3271743>]
- [2] qMiller, J., Dodge, J., & Slavin, R. (2020). Detecting and assessing AI-generated text: Toward a novel toolkit for summarization and paraphrase detection. In Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP) (pp. 6650-6660). [Link][<https://www.aclweb.org/anthology/2020.emnlp-main.540/>]